

Single-Cell Proteomics Accelerates toward Proteoforms

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Analyzing the proteomes of single cells allows for characterizing cellular heterogeneity and may support the inference of regulatory mechanisms for diverse phenotypes.¹ Mass spectrometry (MS) single-cell proteomics is predominantly performed using bottom-up approaches that can quantify peptides from thousands of proteins across many single cells.^{1–3} Yet, bottom-up MS is limited in the identification of combinatorial modifications that contribute to proteoforms. Most gene-templated proteins are already known to contain at least one modification;⁴ these ubiquitous modifications contribute to proteoform diversity and modulate cellular activity. Thus, direct detection and quantification of proteoforms, usually via top-down MS, is a major frontier in single-cell proteomics.

Top-down MS proteomics of single cells confronts many challenges, including limited sensitivity due to the spreading of protein ions across isotopic and charge-state peaks. Successful approaches thus far report dozens of identified proteoforms per single cell using capillary electrophoresis⁵ or liquid chromatography.⁶ Yet, the challenges of multiplexing top down MS limit the throughput of approaches requiring liquid phase separation. Su et al.⁷ circumvent this throughput bottleneck by cleverly combining an established surface-sampling technique, nanospray desorption electrospray ionization (nano-DESI)⁸ with individual ion MS detection (I²MS),^{9,10} which enhances both mass resolution and sensitivity.

The combination of nano-DESI and I²MS by Su et al. demonstrates a promising approach for single-cell proteoform imaging MS (*scPiMS*).⁷ In their approach, cultured cells are drop cast onto a glass slide for analysis, using a fixation procedure optimized for compatibility with nano-DESI. In the targeted *scPiMS* experiment demonstrated, the extraction droplet is parked over individual cells, allowing for desorption of proteins until ion signal diminishes, usually after 10 s. Protein ions are detected by I²MS, which allows for direct determination of their *m/z* and charge to generate a mass spectrum. The method is compatible with an untargeted approach in which the sampling probe is rastered across an entire slide at a constant rate, as demonstrated by the analysis of over 500 single cells in less than 1 day. In this higher throughput approach, 38 proteoforms were identified from a mouse embryonic fibroblast cell line (NIH 3T3 cells). The authors estimate that up to approximately 1,000 cells/day can be analyzed in the current configuration.

scPiMS motivates further research both for increasing the number of detected distinct proteoforms and for increasing the

number of copies sampled from each proteoform in each single cell. Currently, multiple proteoforms detected by *scPiMS* remain unassigned, pointing to the potential to increase proteoform coverage by advancing data interpretation. While I²MS can detect and identify single ions, reliable quantification requires counting multiple ions per proteoform.² Therefore, increasing the number of sampled copies represents an important opportunity for improving the quantitative performance of *scPiMS* and thus enabling reliable proteoform quantification and modeling in single cells.^{1,2}

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